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Due Date: 2/10/26

P.set #: 2

Date Submitted: 2/9/26

#1 **Known:** $R = 0.5\,m$, $H = 1\,m$, $T_{water} = T_s = 50°C$, $T_\infty = 20°C$, $h = 15\,\frac{W}{m^2 \cdot K}$, $L = 2\,cm$, $k = 0.1\,\frac{W}{m \cdot K}$

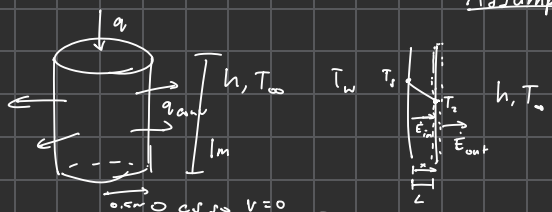
Find: a) q required to maintain $T_s = T_{water} = 50°C$

b) same thing but with insulation foam

c) compare a and b's results

Schematic:

Assumptions: $\dot{E}_{ss} = 0$, $q_{rad} = 0$



0.5m 0, cs so V = 0

Analysis: a) $\dot{E}_{in} + \cancel{\dot{E}_g} - \dot{E}_{out} = \cancel{\dot{E}_{s.s.}}$ (0, 0)

$H \cdot R \cdot 2 \cdot \pi + \pi \cdot R^2$

$\rightarrow q - q_{conv} = 0 \rightarrow q = h \cdot A \cdot (T_s - T_\infty)$

$= (15\,\frac{W}{m^2 \cdot K})(1m \cdot 0.5m \cdot 2 \cdot \pi + \pi \cdot R^2)(50°C - 20°C)$

$$\boxed{q = 1{,}413.72\,W}$$

b) $\dot{E}_{in} + \cancel{\dot{E}_g} - \dot{E}_{out} = \cancel{\dot{E}_{s.s.}}$ (0, cs; 0)

$\rightarrow$ at interface: $\dot{E}_{out} = \dot{E}_{in}$ $\qquad T_\infty = T_2$

$\rightarrow -k \frac{T_2 - T_s}{L} A = hA(T_\infty + T_2)$

$\frac{k}{L} T_s - \frac{k}{L} T_2 = hT_2 - hT_\infty$

$\rightarrow \frac{k}{L} T_s + hT_\infty = T_2(\frac{k}{L} + h)$

$\frac{(\frac{k}{L} T_s + hT_\infty)}{\frac{k}{L} + h} = T_2 = 27.5°C$

$\rightarrow q = q_{conv} = hA(T_2 - T_\infty)$

$= h(\pi R^2 + 2\pi R H)(T_2 - T_\infty)$

$= 15\,\frac{W}{m^2 \cdot K}(\pi(0.5m)^2 + 2\pi(0.5m)(1m))(27.5°C - 20°C)$

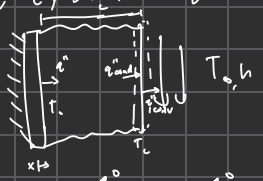
$= \boxed{441.79\,W = q}$

c) The heat rate in required to maintain $T_w = 50°C$ is drastically smaller with insulation than without it, as the heater's surface T is less due to the conduction occuring through the insulation. This decreases convection h.t. and the energy required.

#2 | **Known:** $L = 10\ cm$, $k = 1\ \frac{W}{m \cdot K}$, $T_\infty = 20°C$, $h = 100\ \frac{W}{m^2 \cdot K}$, $q'' = 1000\ \frac{W}{m^2}$

Find: a) T_o, T_c, b) maximum q'' if $T_o = 200°C$, c) three strategies to increase q''_{max} under same max $T = 200°C$

Schematic:



Assumptions: $\dot{E}_{st} = 0$

Analysis: a) $\dot{E}_{in} + \cancel{\dot{E}_g}^{\,0} - \dot{E}_{out} = \cancel{\dot{E}_{st}}^{\,0}$

$\therefore \dot{E}_{in} = \dot{E}_{out}$

$\rightarrow q''_{cond} = q''_{conv} \Rightarrow -k \cdot \frac{T_c - T_o}{L} = h \cdot (T_c - T_\infty) \rightarrow T_o = hL(T_c - T_\infty)k^{-1} + T_c$

$= \left(100\ \frac{W}{m^2 \cdot K}\right)(0.1m)(30°C - 20°C)\left(1\ \frac{W}{m \cdot K}\right)^{-1} = 100°C$

$= T_o$

$\rightarrow q'' = q''_{conv} = q''_{cond} \rightarrow q'' = hT_c - hT_\infty \rightarrow T_c = \frac{q'' + hT_\infty}{h} = \frac{1000\ \frac{W}{m^2}}{100\ \frac{W}{m^2 \cdot K}} + 20°C = \boxed{30°C = T_c}$

b) $q'' = q''_{cond} = -\frac{k(T_c - T_o)}{L}$

aside: $q'' = q''_{conv} = h(T_c - T_\infty)$

$\Rightarrow q'' = -\frac{k}{L}\left(\frac{q''}{h} + T_\infty\right) + \frac{kT_o}{L}$

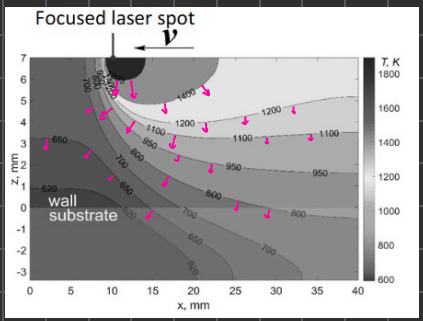
$\rightarrow q''\left(1 + \frac{k}{Lh}\right) = \frac{kT_o}{L} - \frac{kT_\infty}{L}$

$\rightarrow q'' = \frac{1\ \frac{W}{m \cdot K}}{0.1m}(200°C - 20°C)\left(1 + \frac{1\ \frac{W}{m \cdot K}}{0.1m \cdot 100\ \frac{W}{m^2 \cdot K}}\right)^{-1}$

$= \boxed{1636\ \frac{W}{m^2} = q''}$

c) The final equation from part (b) shows that increasing T_∞, h, or L will all increase the heat flux without adjusting the temperature. Using a coolant with a higher heat transfer coefficient, increasing the coolant's temperature, or increasing the thickness of the plastic will all increase the maximum allowable heat flux.

3 a)



All pink arrows (↑) represent direction of heat flux. They are normal to an isotherm, increasing in magnitude (length of arrow) with increasing isotherm density, meaning a greater temperature gradient.

b) The left side has a greater heat flux, as the conductive heat equation in 2D tells us that heat flux increases with increasing temperature gradient. The left side's comparably close isotherms indicate a higher temperature gradient, and thus higher heat flux than the right side.